Statistical methods for inferring the history of languages

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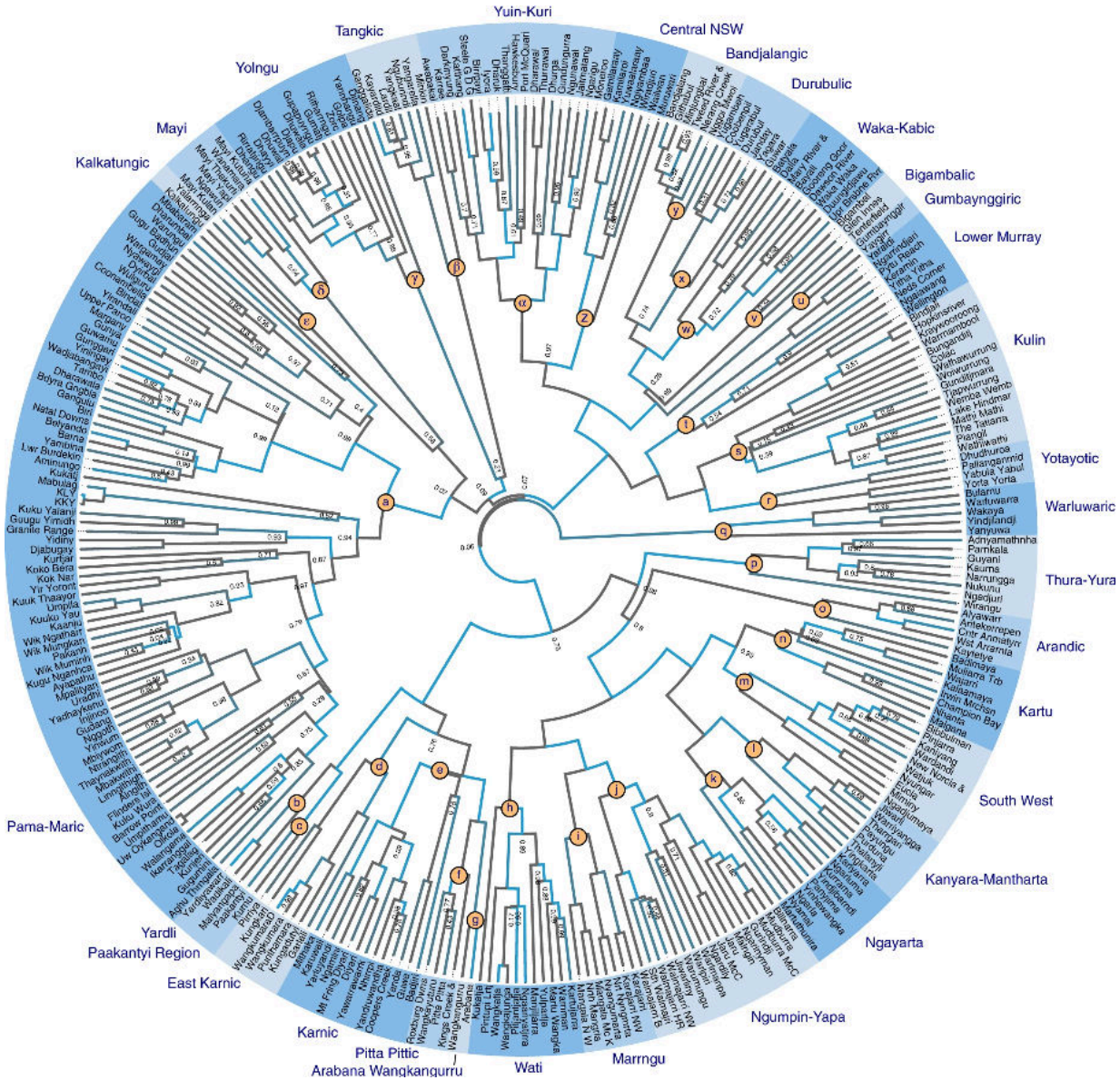
15 October 2026
Interactions between Formal and Computational Linguistics seminar

Based on work with (a) Erich Round and Rikker Dockum, (b) Guillaume Jacques, Laurent Sagart, Yunfan Lai, Valentin Thouzeau, Simon Greenhill and Mattis List, (c) Grégoire Clarté, Natasha Abner, Carlo Geraci, Justine Mertz, Anah Salgat and Shi Yu, (d) Ambre Salis and Axel Molina, (e) Christoph Grüter and Lucie Ravaux, and (f) Valentin Thouzeau, Guillaume Dezecache, Ewan Dunbar, Philippe Schlenker and Emmanuel Chemla.

Phylolinguistics

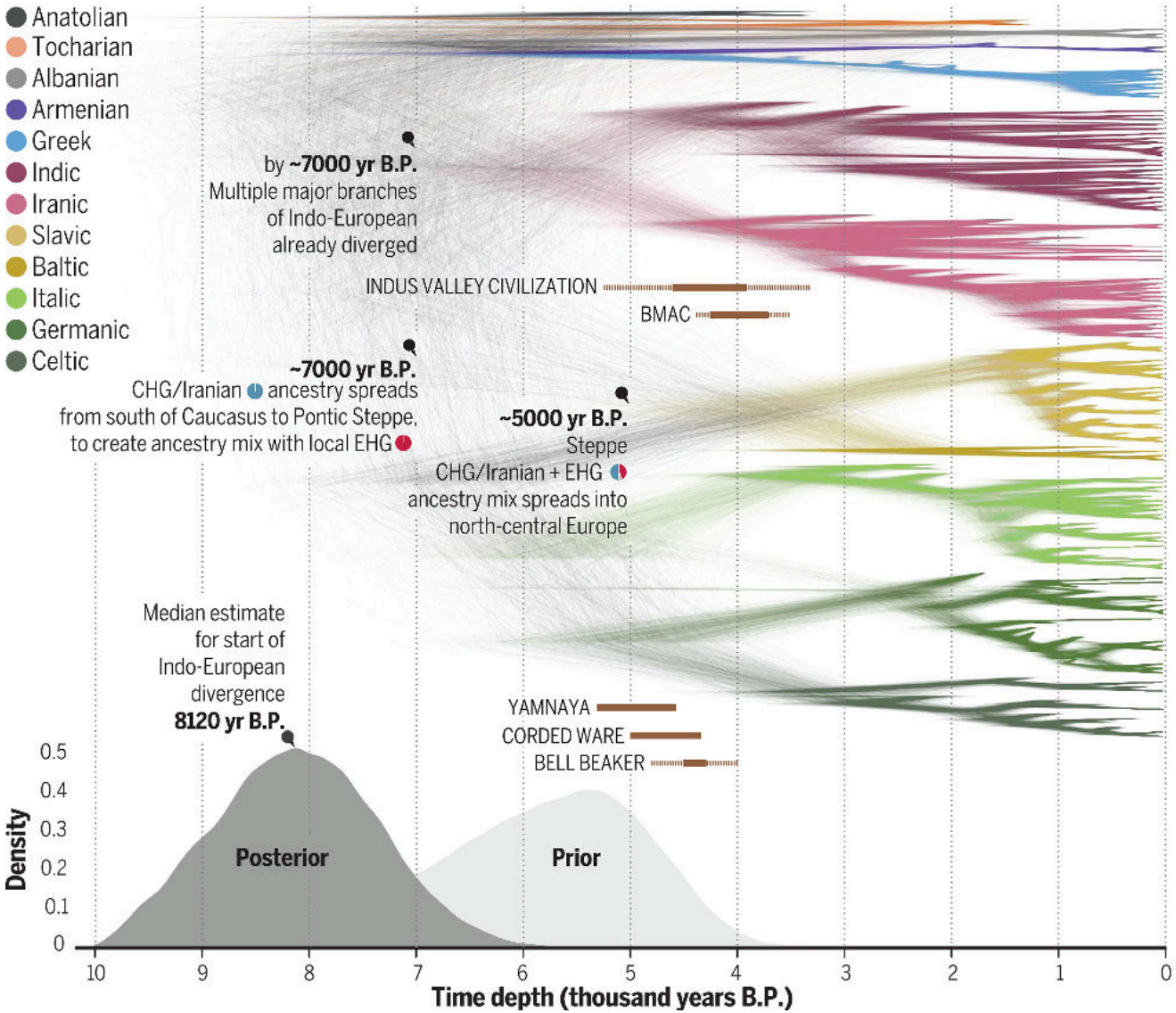
- Phylolinguistics is a collection of statistical methods to reconstruct the history of a family of languages, based on data observed at modern or recent languages
- Most standard model: tree
- Traditionally reconstructed by hand using the comparative method
- Quantitative methods have become very popular since 2003, with several high-profile studies

Inferring trees



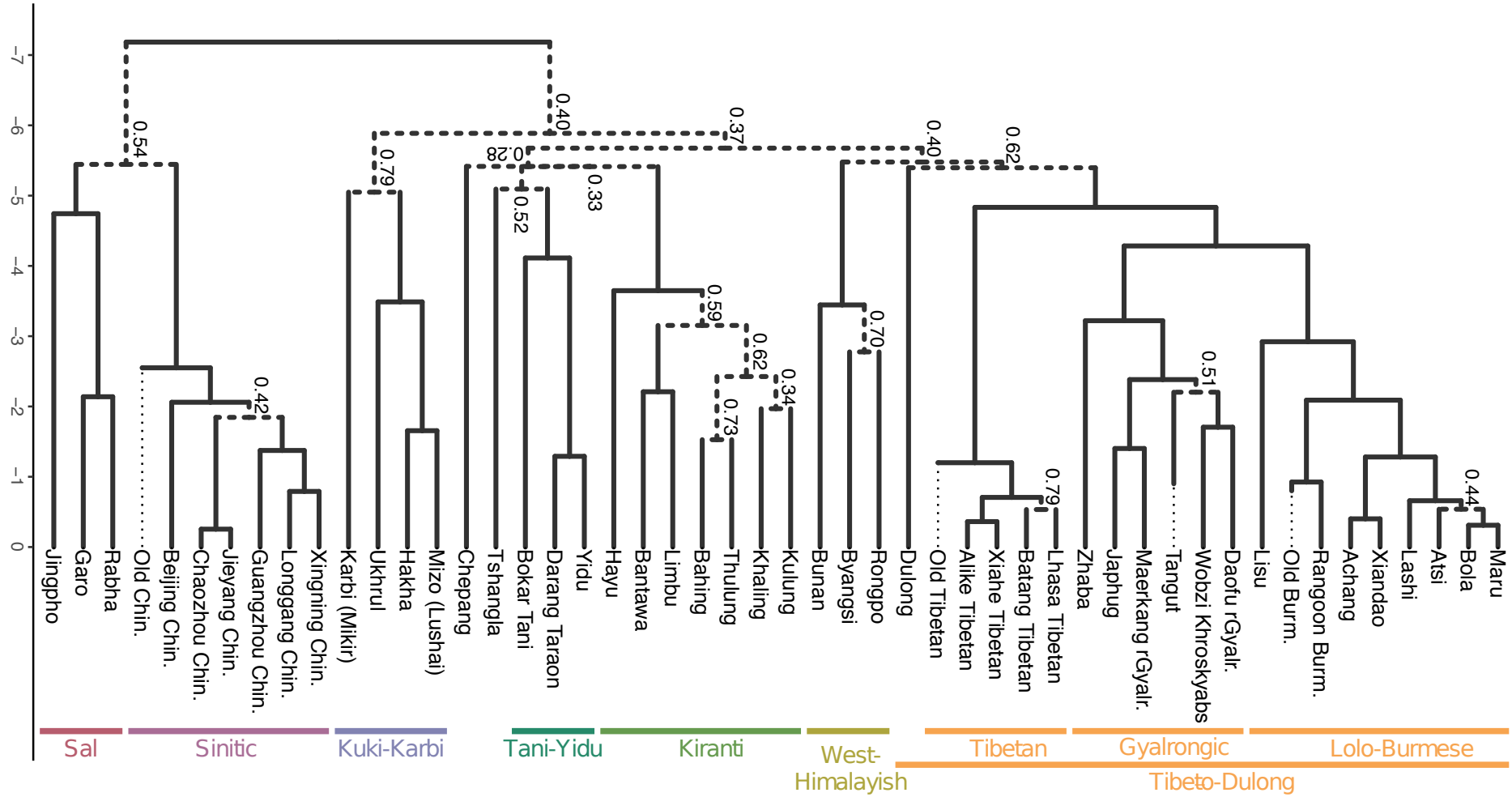
Pama-Nyungan, Bouckaert et al. (2018), Nature E&E

Inferring trees



Indo-European, Heggarty et al. (2023), Science

Inferring trees



Sino-Tibetan, Sagart et al. (2019), PNAS

Context

Over the past 20 years, a large number of recent papers have described computationally-intensive statistical methods for Historical Linguistics, due to:

- Increased computational power
- Advances in statistical methodology
- New datasets
- Complex linguistic questions which cannot be answered with traditional methods

Questions of interest

- Which languages are related?
- Given a set of related languages, can we reconstruct their history and the age of the most recent common ancestor (MRCA)?
- What mechanisms drive language change?
- How do the various parts of language change? Vocabulary, syntax, phonetics...

Statistical workflow in a nutshell

1. Collect data
2. Design model
3. Perform inference (MCMC, ...)
4. Check convergence
5. In-model validation (is our inference method able to answer questions from our model?)
6. Model mis-specification analysis (do we need a more complex model?)
7. Conclude

In general, it is more difficult to perform inference for a more complex model.

See also Gelman et al. (2020) on the Bayesian workflow.

Outline

1. Swadesh: Glottochronology
2. Gray & Atkinson: Language phylogenies
3. Sagart et al.: Sino-Tibetan phylogenies
4. Clarté & Ryder + Abner et al.: Lexicon and Phonology modelled jointly
5. Evolution of animal calls
6. Conclusions

LEXICO-STATISTIC DATING OF PREHISTORIC ETHNIC CONTACTS

With Special Reference to North American Indians and Eskimos

MORRIS SWADESH

PREHISTORY refers to the long period of early human society before writing was available for the recording of events. In a few places it gives way to the modern epoch of recorded history as much as six or eight thousand years ago; in many areas this happened only in the last few centuries. Everywhere prehistory represents a great obscure depth which science seeks to penetrate. And indeed powerful means have been found for illuminating the unrecorded past, including the evidence of archeological finds and that of the geographic distribution of cultural facts in the earliest known periods. Much depends on the painstaking analysis and comparison of data, and on the effective reading of their implications. Very important is the combined use of all the evidence, linguistic and ethnographic as well as archeological, biological, and geological. And it is essential constantly to seek new means of expanding and rendering more accurate our deductions about prehistory.

One of the most significant recent trends in the

measuring the amount of radioactivity still going on. Consequently, it is possible to determine within certain limits of accuracy the time depth of any archeological site which contains a suitable bit of bone, wood, grass, or any other organic substance.

Lexicostatistic dating makes use of very different material from carbon dating, but the broad theoretical principle is similar. Researches by the present author and several other scholars within the last few years have revealed that the fundamental everyday vocabulary of any language—as against the specialized or "cultural" vocabulary—changes at a relatively constant rate. The percentage of retained elements in a suitable test vocabulary therefore indicates the elapsed time. Wherever a speech community comes to be divided into two or more parts so that linguistic change goes separate ways in each of the new speech communities, the percentage of common retained vo-

First attempt: Swadesh (1952)

Aim: dating the MRCA (Most Recent Common Ancestor) of a pair of languages.
Data: "core vocabulary" (Swadesh lists). 215 or 100 words.

Core vocabulary



Assumptions

Swadesh assumed that core vocabulary evolves at a constant rate (through time, space and meanings). Given a pair of languages with percentage C of shared cognates, and a constant retention rate r, the age t of the MRCA is

$$t = \frac{\log C}{2 \log r}$$

The constant r was estimated using a pair of languages for which the age of the MRCA is known.

Issues with glottochronology

Many statistical shortcomings. Mainly:

1. Simplistic model
2. No evaluation of uncertainty of estimates
3. Only small amounts of data are used

Bergsland and Vogt (1962) debunked glottochronology, showing on 3 pairs of languages with known history that the assumption of constant rates does not hold.

What has changed?

- More elaborate models + model misspecification analyses
- We can estimate the uncertainty ($\Rightarrow$ easier to answer "I don't know")
- Large amounts of data

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rheology of the ascending magma, our findings are in a broader sense equivalent to Eichelberger's hypothesis[1] that "higher viscosity of magma may favour non-explosive degassing rather than hinder it", albeit with the added complexity of shear-induced fragmentation. □

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Language-tree divergence times support the Anatolian theory of Indo-European origin

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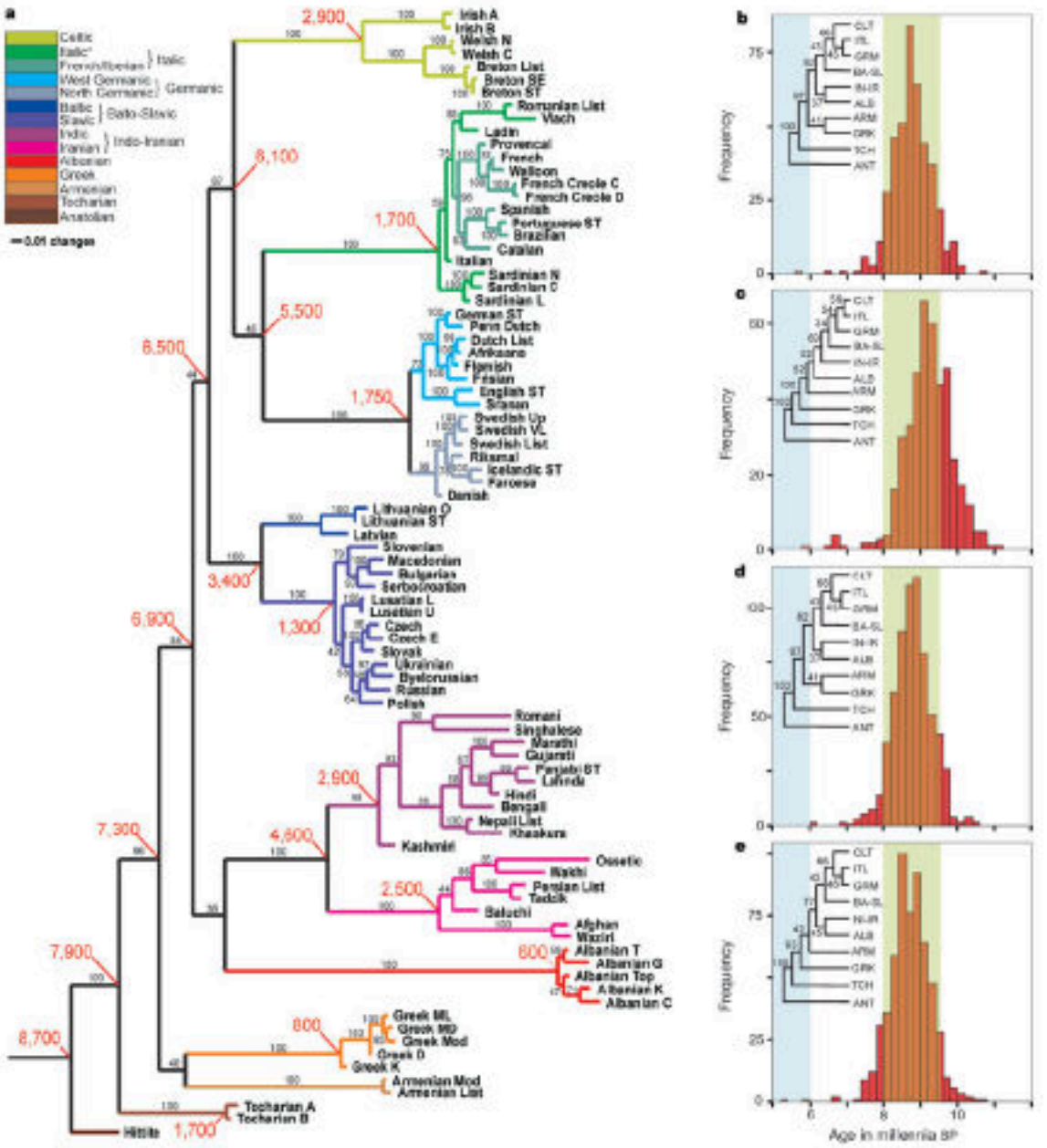
Languages, like genes, provide vital clues about human history[1,2]. The origin of the Indo-European language family is "the most intensively studied, yet still most recalcitrant, problem of historical linguistics"[3]. Numerous genetic studies of Indo-European origins have also produced inconclusive results[4,5,6]. Here we analyse linguistic data using computational methods derived from evolutionary biology. We test two theories of Indo-European origin: the 'Kurgan expansion' and the 'Anatolian farming' hypotheses. The Kurgan theory centres on possible archaeological evidence for an expansion into Europe and the Near East by Kurgan horsemen beginning in the sixth millennium BP[7,8]. In contrast, the Anatolian theory claims that Indo-European languages expanded with the spread of agriculture from Anatolia around 8,000–9,500 years BP[9]. In striking agreement with the Anatolian hypothesis, our analysis of a matrix of 87 languages with 2,449 lexical items produced an estimated age range for the initial Indo-European divergence of between 7,800 and 9,800 years BP. These results were robust to changes in coding procedures, calibration points, rooting of the trees and priors in the bayesian analysis.

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Swadesh lists, better analysed

- Use Swadesh lists for 87 Indo-European languages, and a phylogenetic model from Genetics
- Assume a tree-like model of evolution with constant rate of change
- Bayesian inference via MCMC (Markov Chain Monte Carlo)
- Reconstruct trees and dates
- Main parameter of interest: age of the root (Proto-Indo-European, PIE)

Lexical trees



Correcting the issues with glottochronology

Returning to the issues with Swadesh's glottochronology:

1. Simplistic model → Slightly better, but the model of evolution is rudimentary
2. No evaluation of uncertainty of estimates → Bayesian inference
3. Only small amounts of data are used → Large number of languages reduces variability of estimates

Bayesian inference for lexical trees

- The tree parameter is seen as random: it has a distribution
- Via MCMC, G & A get a sample of possible trees, with associated probabilities, rather than a single tree
- The uncertainty in trees is thus made explicit

G & A: conclusions

- Age of PIE: 7800-9800 BP (Before Present)
- Large error bars, but this is a good thing
- Reconstruct many known features of the tree of Indo-European languages
- Little validation of the model, no model misspecification analysis
- More sophisticated models have since been applied to these data, with extensive validation studies.
- These trees can also be used as a building block to answer other questions.

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Sagart et al. (2019)

PNAS

Dated language phylogenies shed light on the ancestry of Sino-Tibetan

Laurent Sagart[a,1], Guillaume Jacques[a,1], Yunfan Lai[b], Robin J. Ryder[c], Valentin Thouzeau[c], Simon J. Greenhill[b,d], and Johann-Mattis List[b,2]

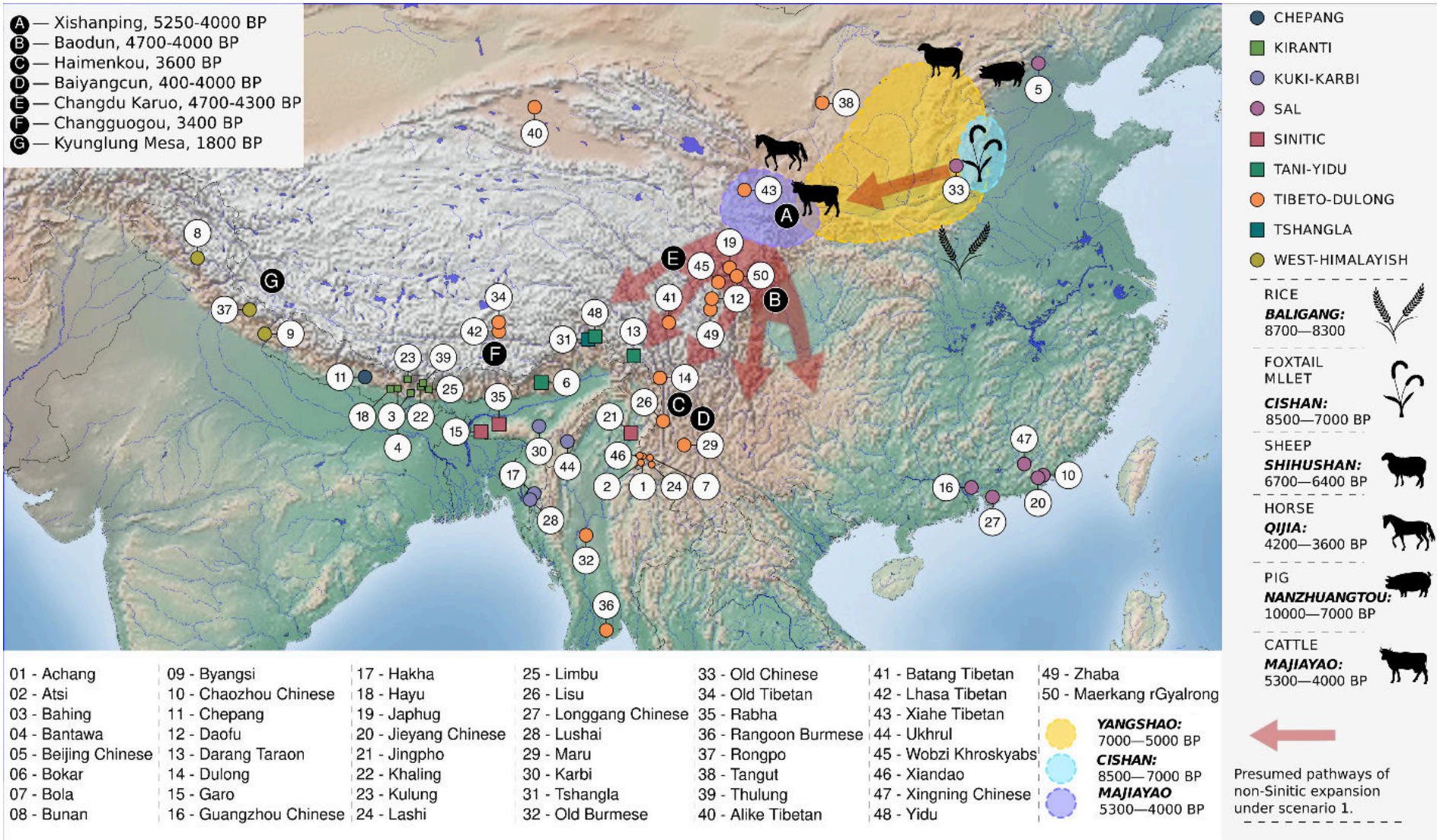
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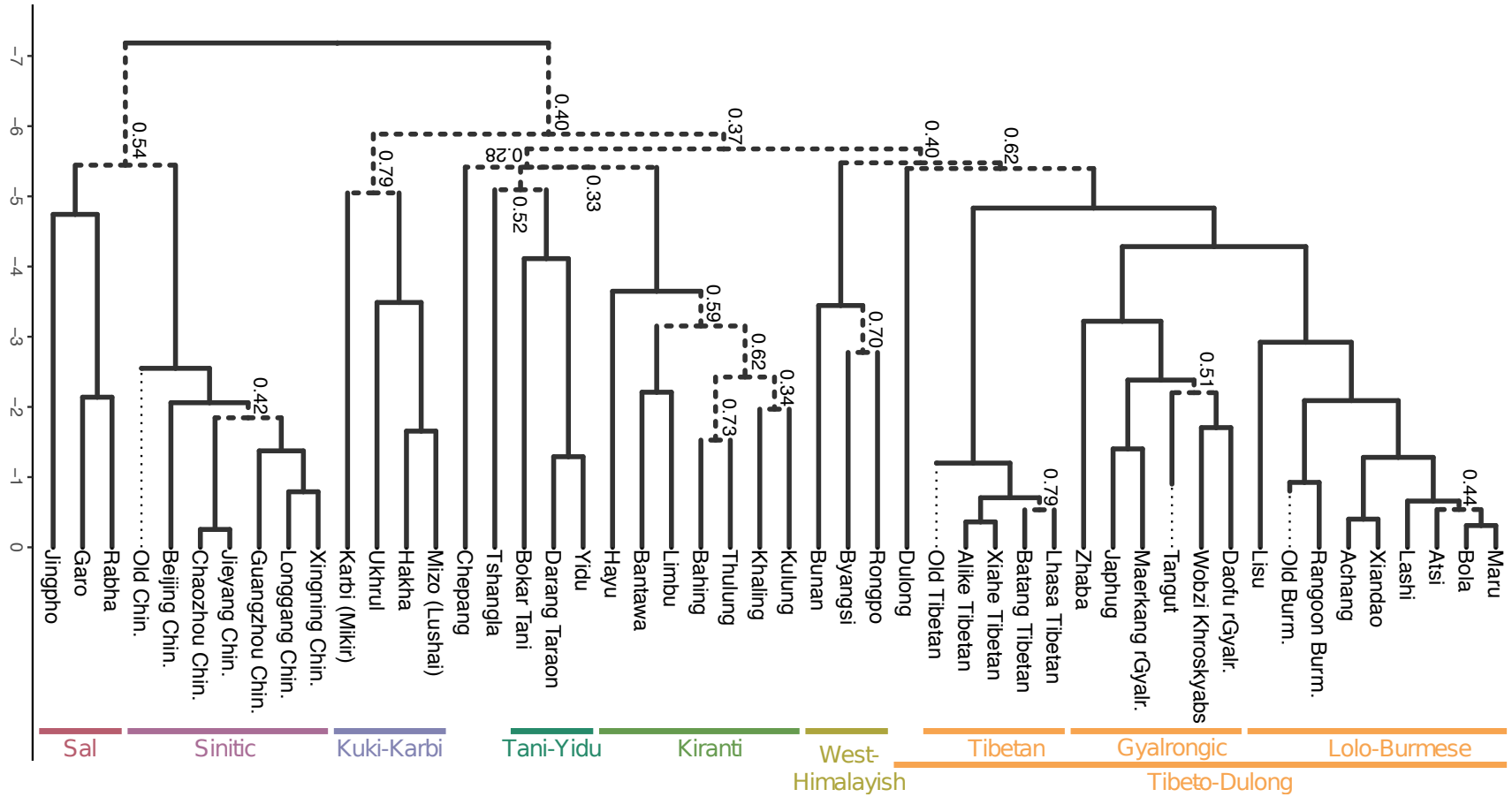
The Sino-Tibetan language family is one of the world's largest and most prominent families, spoken by nearly 1.4 billion people. Despite the importance of the Sino-Tibetan languages, their prehistory remains controversial, with ongoing debate about when and where they originated. To shed light on this debate we develop a database of comparative linguistic data, and apply the linguistic comparative method to identify sound correspondences and establish cognates. We then use phylogenetic methods to infer the relationships among these languages and estimate the age of their origin and homeland. Our findings point to Sino-Tibetan originating with north Chinese millet farmers around 7200 B.P. and suggest a link to the late Cishan and the early Yangshao cultures.

tions in Chinese date to before 1400 BCE, and Chinese has an abundant and well-studied literature dating back to the early first millennium BCE. The Shāng Kingdom, the Chinese polity associated with these inscriptions, was centered on the lower Yellow River valley. Gradual annexation of neighboring regions and shift of their peoples to the Chinese language led to the striking numerical predominance of Chinese speakers today, and, consequently, to the lack of linguistic diversity in the eastern part of the Sino-Tibetan domain. Tibetan, Tangut, Newar, and Burmese, the family's other early literary languages, were reduced to script considerably more recently: The oldest texts in these languages date from 764 CE, 1070 CE, 1114 CE, and 1113 CE, respectively. The area with the most diverse Sino-Tibetan languages

Sino-Tibetan languages



Example of a (consensus) tree



Questions to answer

- Topology of the tree: subfamilies and their links
- Age of ancestor nodes
- Age of root

Swadesh lists

- 100 or 200 words, present in almost all languages: *bird, hand, to eat, red*...
- Cognacy judgments performed by experts
- "Obvious" borrowings removed

Data are transformed to a binary matrix. This observation process is the first aspect we need to model.

Binary data: *he dies, three, all* (data: Ringe et al. '02)

	he dies	*three*	*all*
Old English	stierfþ	þrīe	ealle
Old High German	stirbit, touwit	drī	alle
Avestan	miriiete	þrāiiō	vispe
Old Church Slavonic	umĭretŭ	trĭje	vĭsi
Latin	moritur	trēs	omnēs
Oscan	?	trís	súllus

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Cognacy classes (traits) for the meaning *he dies*:

1. {stierfþ, stirbit}
2. {touwit}
3. {miriiete, umĭretŭ, moritur}

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O. English	1	0	0
OH German	1	1	0
Avestan	0	0	1
OC Slavonic	0	0	1
Latin	0	0	1
Oscan	?	?	?

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OH German	1	1	0	1
Avestan	0	0	1	1
OC Slavonic	0	0	1	1
Latin	0	0	1	1
Oscan	?	?	?	1

Cognacy classes for the meaning *three*:

1. {þrīe, drī,þrāiiō, trĭje, trēs, trís}

Binary data: *he dies, three, all* (data: Ringe et al. '02)

	he dies	*three*	*all*
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O. English	1	0	0	1	1	0	0	0
OH German	1	1	0	1	1	0	0	0
Avestan	0	0	1	1	0	1	0	0
OC Slavonic	0	0	1	1	0	1	0	0
Latin	0	0	1	1	0	0	1	0
Oscan	?	?	?	1	0	0	0	1

Cognacy classes for *all*:

1. {ealle, alle}
2. {vispe, vĭsi}
3. {omnēs}
4. {súllus}

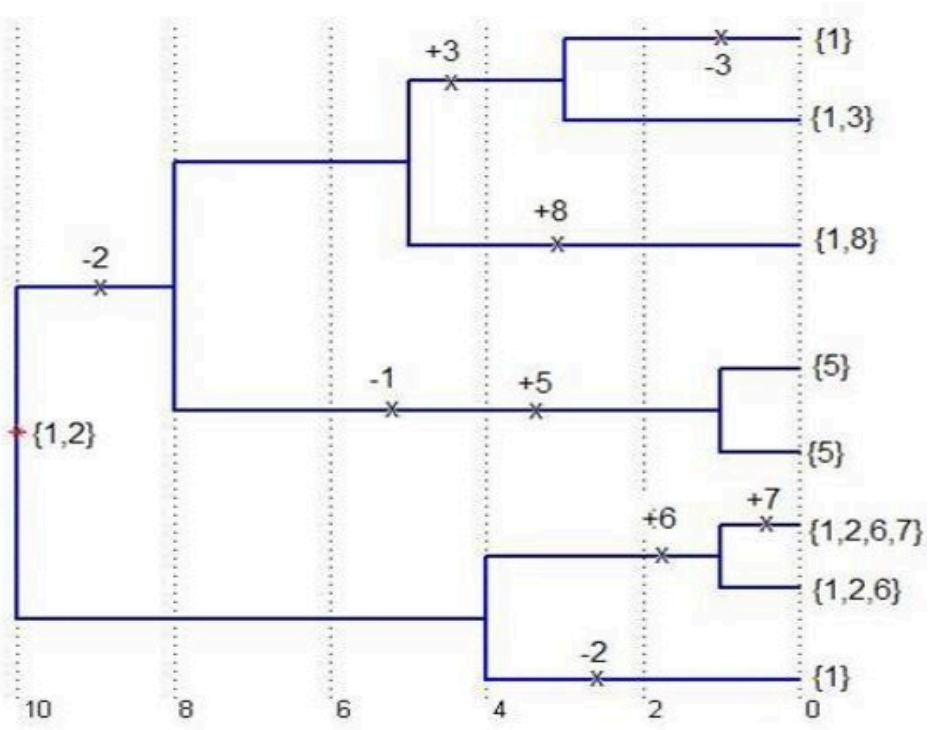
Final data

Old English	1	0	0	1	1	0	0	0
Old High German	1	1	0	1	1	0	0	0
Avestan	0	0	1	1	0	1	0	0
Old Church Slavonic	0	0	1	1	0	1	0	0
Latin	0	0	1	1	0	0	1	0
Oscan	?	?	?	1	0	0	0	1

Constraints

- Constraints on the tree topology
- Constraints on the age of some nodes or ancient languages
- These constraints are used to estimate the evolution rates and the age.
- Also provide one way of validating the model and inference procedure.

Model (1): birth-death process



- Traits (=cognacy classes) are born at rate λ.
- Traits die at rate μ.
- λ and μ are constant.

1	1	0	0	0	0	0	0	0
2	1	0	1	0	0	0	0	0
3	1	0	0	0	0	0	0	1
4	0	0	0	0	1	0	0	0
5	0	0	0	0	1	0	0	0
6	1	1	0	0	0	1	1	0
7	1	1	0	0	0	1	0	0
8	1	0	0	0	0	0	0	0

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In general, it is more difficult to perform inference for a more complex model.

See also Gelman et al. (2020) on the Bayesian workflow.

Inference

- BEAST and TraitLab software
- Bayesian inference
- Markov Chain Monte Carlo
- (Almost) uniform prior over the age of the root
- Extensive validation (in-model and out-model; real data and synthetic data)

Posterior distribution

$$
\begin{aligned}
& p(g, \mu, \lambda, \kappa, \rho, \xi | \mathbf{D} = D) \\
&= \frac{1}{N!} \frac{\lambda^N}{\mu^N} \exp\left(-\frac{\lambda}{\mu} \sum_{\langle i,j \rangle \in E} P[\mathcal{E}_Z | Z = (t_i, i), g, \mu, \kappa, \xi](1 - e^{-\mu(t_j - t_i + k_i T_C)}) \right) \\
&\quad \times \prod_{a=1}^{N} \left(\sum_{\langle i,j \rangle \in E_a} \sum_{\omega \in \Omega_a} P[M = \omega | Z = (t_i, i), g, \mu](1 - e^{-\mu(t_j - t_i + k_i T_C)}) \right) \\
&\quad \times \frac{1}{\mu\lambda} p(\rho) f_G(g|T) \frac{e^{-\rho|g|}(\rho|g|)^{k_T}}{k_T!} \prod_{i=1}^{L} (1 - \xi_i)^{Q_i} \xi_i^{N - Q_i}
\end{aligned}
$$

Likelihood calculation

$$\sum_{\omega \in \Omega_a^{(c)}} P[M = \omega | Z = (t_i, c), g, \mu] =$$

$$\begin{cases} \delta_{i,c} \times \sum_{\omega \in \Omega_a^{(c)}} P[M = \omega | Z = (t_c, c), g, \mu] & \text{if } Y(\Omega_a^{(c)}) \geq 1 \\ (1 - \delta_{i,c}) + \delta_{i,c} v_c^{(0)} & \text{if } Y(\Omega_a^{(c)}) Q(\Omega_a^{(c)}) = 0 \\ & \quad (\textit{i.e. } \Omega_a^{(c)} = \{\varnothing\}) \\ 1 - \delta_{i,c}(1 - \sum_{\omega \in \Omega_a^{(c)}} P[M = \omega | Z = (t_c, c), g, \mu]) & \text{if } Y(\Omega_a^{(c)}) = 0 \\ & \text{and } Q(\Omega_a^{(c)}) \geq 1 \end{cases}$$

$$\sum_{\omega \in \Omega_a^{(c)}} P[M = \omega | Z = (t_c, c), g, \mu] = \begin{cases} 1 & \text{if } \Omega_a^{(c)} = \{\{c\}, \varnothing\} \text{ or } \{\{c\}\} \\ & \quad (\textit{i.e. } D_{c,a} \in \{?, 1\}) \\ 0 & \text{if } \Omega_a^{(c)} = \{\varnothing\} \ (\textit{i.e. } D_{c,a} = 0) \end{cases}$$

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Limitations of this model

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1. Constant rates across time and space
2. No handling of missing data
3. No handling of borrowing
4. Treats all traits in the same fashion
5. Binary coding loses part of the structure
6. Assumes a tree structure
7. ...

Do any of these limitations introduce systematic bias?

Limitations of this model

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4. Treats all traits in the same fashion
5. Binary coding loses part of the structure
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7. ...

Do any of these limitations introduce systematic bias? (Answer: YES, some do.)
Check each misspecification in turn, and adapt the model if necessary.

How to check for misspecifications

1. Compute the Bayes factor to choose between two models $\mathcal{M}_1$ and $\mathcal{M}_2$ (gold standard, but often mathematically challenging and computationally demanding)
2. If a misspecification can be represented by a single estimable parameter θ, estimate it and check whether $\theta = 0$
3. Perform a simulation study: synthesize data from the complex model, infer parameters using the simple model, and check whether we are able to reconstruct the "truth"
4. If for some reason two reasonable models cannot be compared using the above: infer under both, and see where the output agrees.

Influence of borrowing (1)

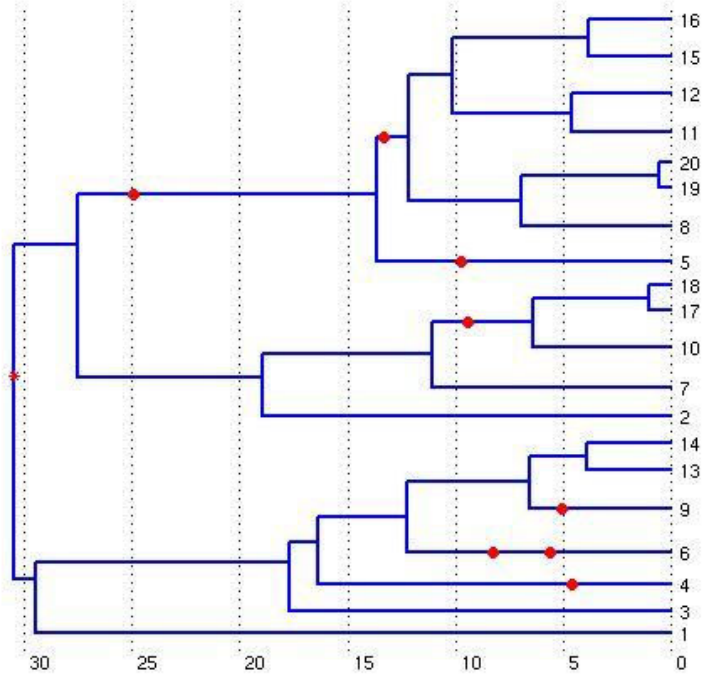


Figure: True tree, 40 words/language, 10% borrowing

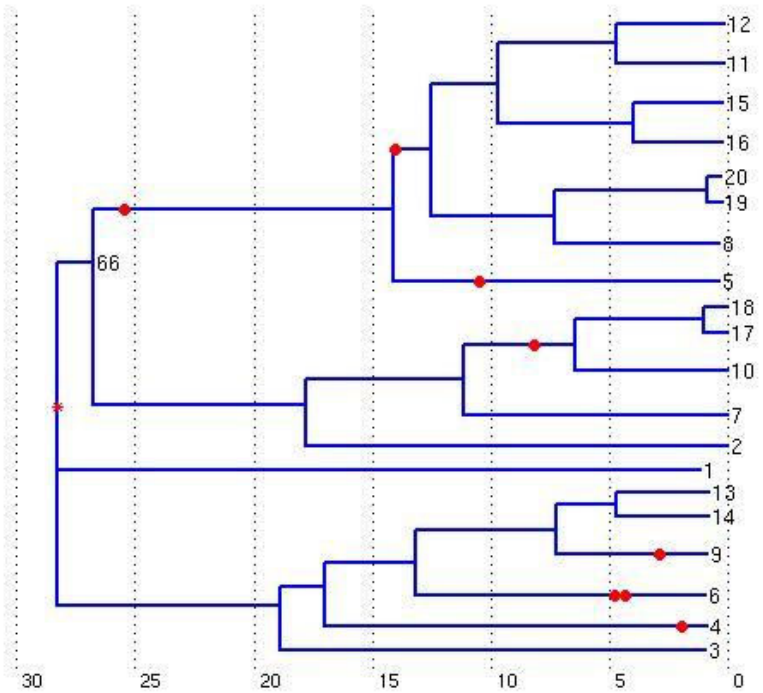


Figure: Consensus tree

With out-of-model synthetic data with low levels of borrowing, the tree is well reconstructed.

Influence of borrowing (2)

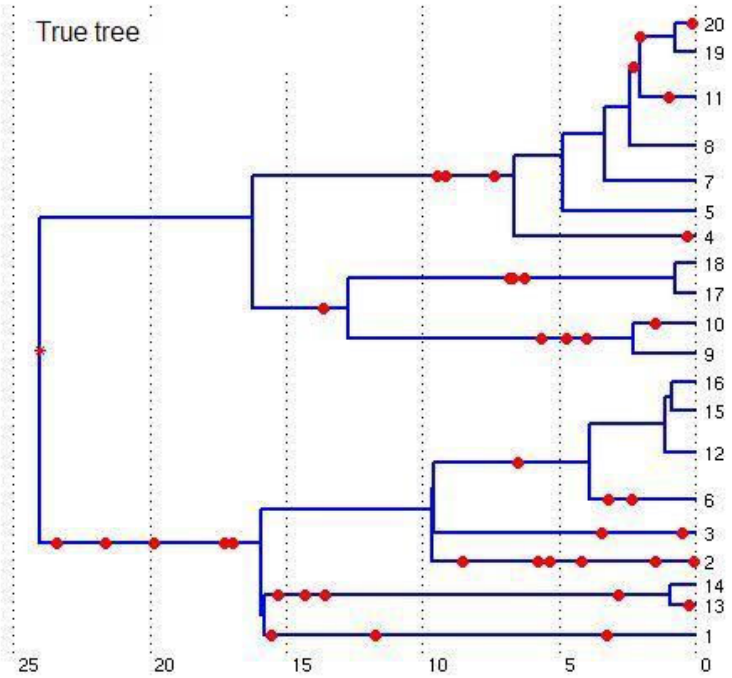


Figure: True tree, 40 words/language, 50% borrowing

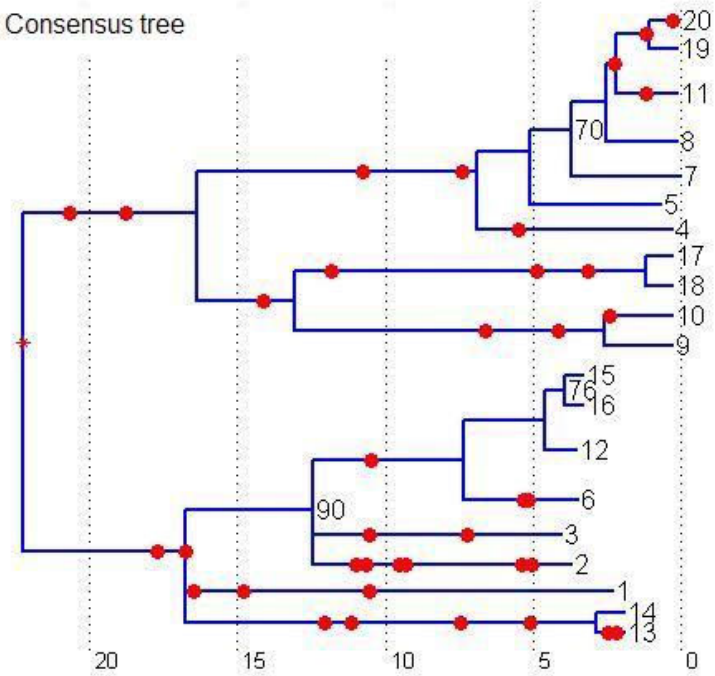


Figure: Consensus tree

Influence of borrowing (3)

- The topology is well reconstructed
- Dates are under-estimated if borrowing levels are high

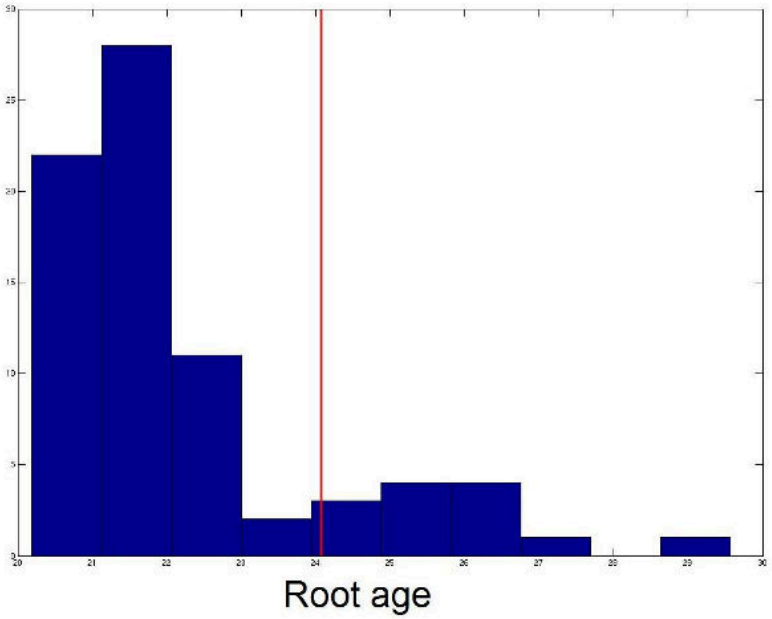


Figure: Root age

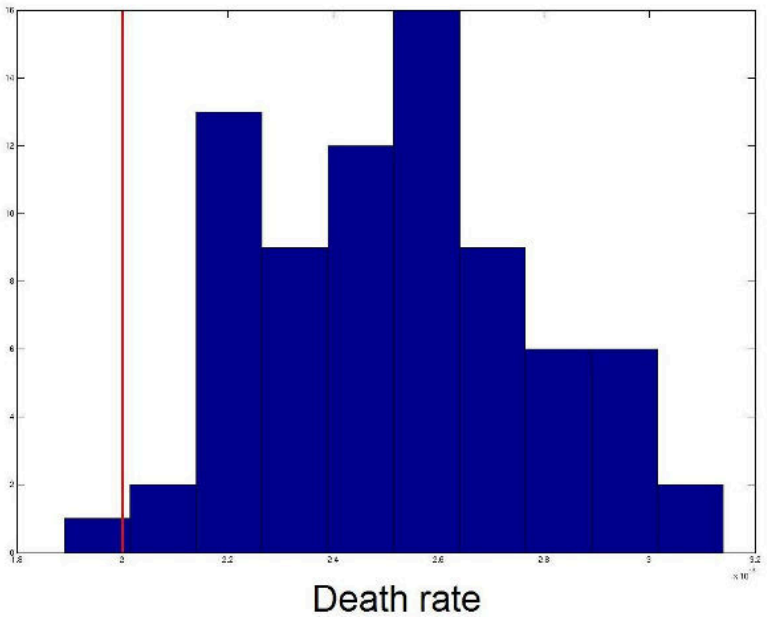


Figure: Death rate (μ)

Borrowing: what to do?

High levels of undetected borrowing (or other non-treeness) would introduce bias in our results.
Fortunately, Kelly & Nicholls (2017) provide a methodology to:

- Essentially infer a network superimposed on a tree
- Test for treeness (check 1)
- Estimate levels of borrowing (allowing check 2 and check 3)

Borrowing: what to do?

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- Essentially infer a network superimposed on a tree
- Test for treeness (check 1)
- Estimate levels of borrowing (allowing check 2 and check 3)

Here:

- log Bayes factor is non-decisive at 1.8, indicating that including the network does not improve the model fit
- the level of borrowing is estimated at $\hat{\beta}/\mu = 0.104$, a level at which we have no systematic bias with synthetic data

Analysis restricted to 15 languages (chosen randomly across subfamilies) for computational reasons. Took 83 hours on 8 cores.

Mis-specifications

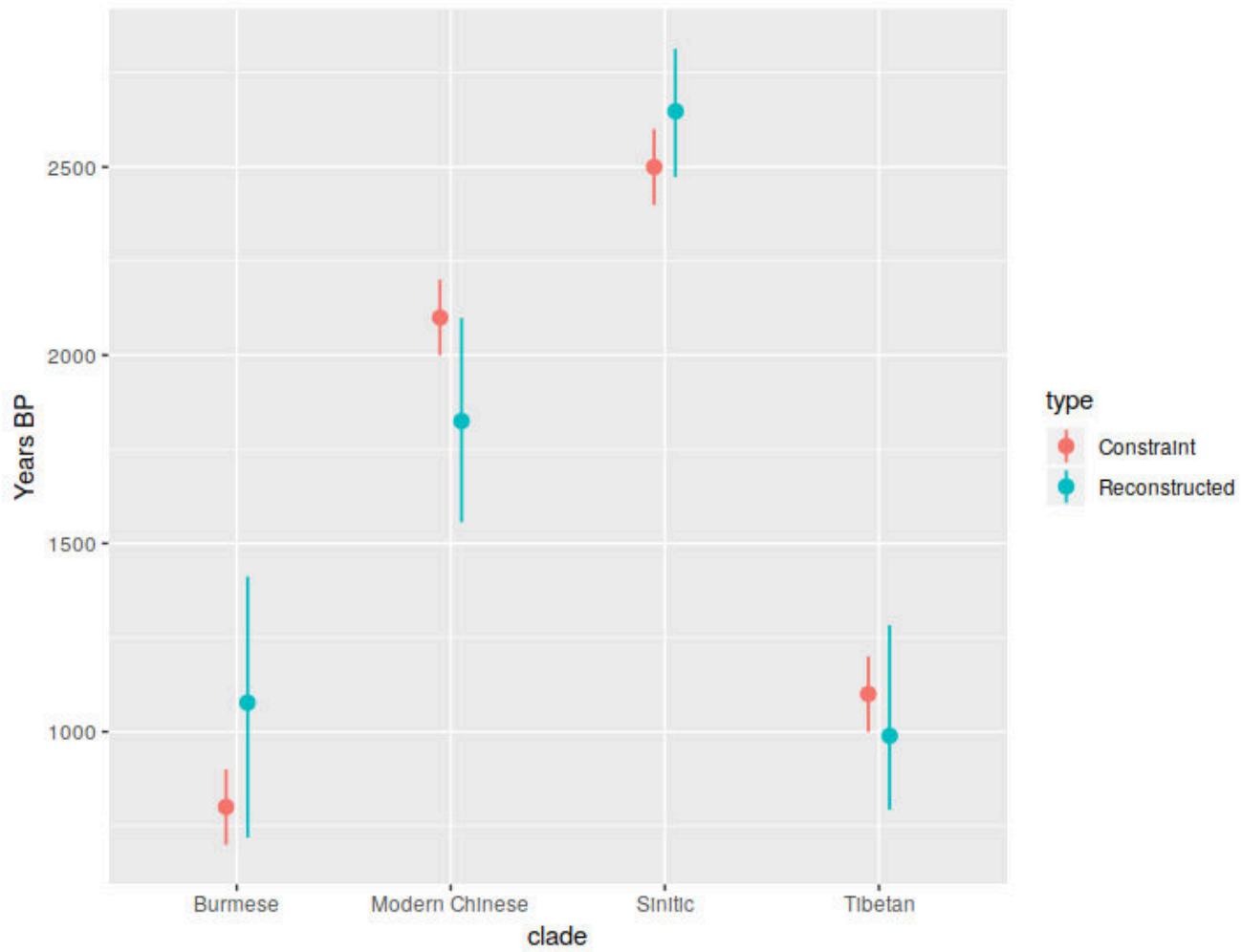
Heterogeneity between traits	Analyse subset of data+ simulated data
Heterogeneity in time/space (non catastrophic)	Infer from 3 distinct models, giving similar results
Borrowing	Bayes factor + Simulated data analysis + check level of borrowing
Data missing in blocks	Simulated data analysis
Non-empty meaning categories	Simulated data analysis
Heterogeneity across subfamilies	Analyse subset of the data
...	...

Cross-validation

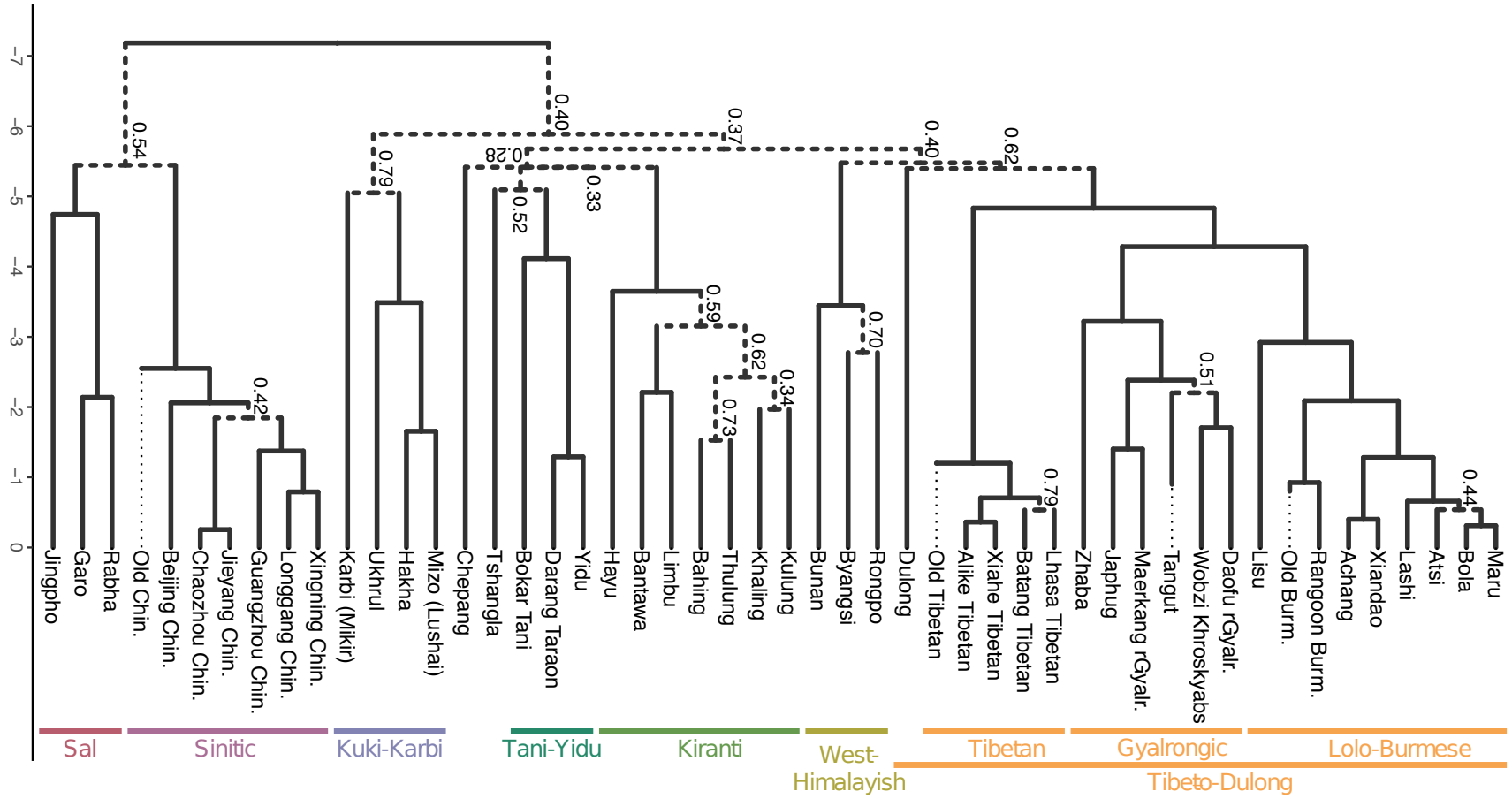
Final check: we can take out the age constraints one by one, and check whether we are able to reconstruct them.

Cross-validation

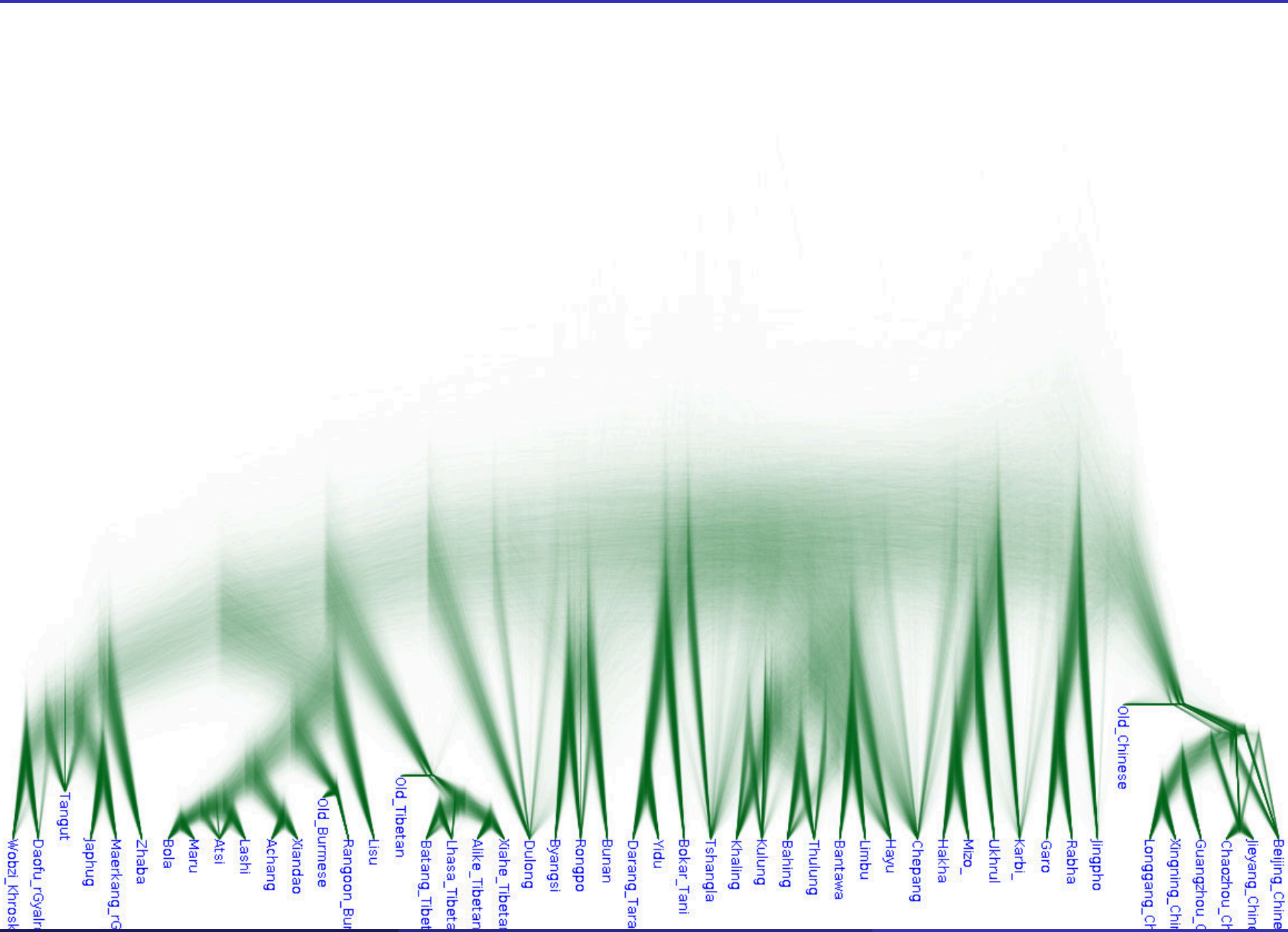
Final check: we can take out the age constraints one by one, and check whether we are able to reconstruct them.



Sino-Tibetan consensus tree



Densitree


Old_Chinese
Beijing_Chine
Jieyang_Chine
Chaozhou_Ch
Guangzhou_C
Xingning_Chi
Longgang_Ch
Jingpho
Rabha
Garo
Karbi_
Ukhrul
Mizo_
Hakha
Chepang
Hayu
Limbu
Bantawa
Thulung
Bahing
Kulung
Khaling
Tshangla
Bokar_Tani
Yidu
Darang_Tara
Bunan
Rongpo
Byangsi
Dulong
Xiahe_Tibetan
Alike_Tibetan
Lhasa_Tibeta
Batang_Tibet
Old_Tibetan
Lisu
Rangoon_Bur
Old_Burmese
Xiandao
Achang
Lashi
Atsi
Maru
Bola
Zhaba
Maerkang_rG
Japhug
Tangut
Daofu_rGyalr
Wobzi_Khrosk

Conclusions

- Root age around 7200 BP (95% HPD: 5100 – 9600 BP)
- Two main expansion scenarios supported: from northern China, or (less probable) from West Himalayas
- Statistics reconstruct known linguistic facts and answer unresolved questions.
- Importance of being Bayesian: uncertainty measured and integrated out; complex model.

Outline

1. Swadesh: Glottochronology
2. Gray & Atkinson: Language phylogenies
3. Sagart et al.: Sino-Tibetan phylogenies
4. Clarté & Ryder + Abner et al.: Lexicon and Phonology modelled jointly
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Abner et al. (2024); Clarté & Ryder (2026) ;

LINGUISTICS

Computational phylogenetics reveal histories of sign languages

Natasha Abner[1]*, Grégoire Clarté[2,3,4]*, Carlo Geraci[5]*, Robin J. Ryder[4]*, Justine Mertz[5,6], Anah Salgat[1,7], Shi Yu[5,8]

Sign languages are naturally occurring languages. As such, their emergence and spread reflect the histories of their communities. However, limitations in historical recordkeeping and linguistic documentation have hindered the diachronic analysis of sign languages. In this work, we used computational phylogenetic methods to study family structure among 19 sign languages from deaf communities worldwide. We used phonologically coded lexical data from contemporary languages to infer relatedness and suggest that these methods can help study regular form changes in sign languages. The inferred trees are consistent in key respects with known historical information but challenge certain assumed groupings and surpass analyses made available by traditional methods. Moreover, the phylogenetic inferences are not reducible to geographic distribution but do affirm the importance of geopolitical forces in the histories of human languages.

Language is a defining feature of human existence, but it is not a static one. Lan- so, uses data from sign languages as an empirical basis for advancing computational

All of the papers presented so far rely on cognacy judgements. Making these judgements often involves reconstructing by hand systematic phonological changes (e.g. $/p/ \rightarrow /f/$).
Very recent work: learning jointly lexical and phonological innovations.

Polynesian data

meaning	*eye*	*seven*	*hand*	*flower*	*to breathe*	*fruit*
Samoan	mata	fitu	lima	fuŋa	mānava	fua
Tahitian	mata	hitu	rima	pua	huti	hua
Hawaiian	maka	hiku	rima	pua	hanu	hua

Polynesian data

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Polynesian data: cognates

	eye_1	$seven_1$	$hand_1$	$flower_1$	$flower_2$	$breathe_1$	$breathe_2$	$breathe_3$	$fruit_1$
Samoan	1	1	1	1	0	1	0	0	1
Tahitian	1	1	1	0	1	0	1	0	1
Hawaiian	1	1	1	0	1	0	0	1	1

Issues with this transformation

- Time consuming
- Requires expert knowledge of the languages
- Cognates can be hard to define/detect
- Deletes information

This is especially problematic for language families where (1) defining cognates is difficult or (2) there have been few lexical innovations, so that most of the signal is in the phonology.

Matrix form

Samoan

eye	m	a	t	a
seven	f	i	t	u
hand	l	i	m	a

Tahitian

eye	m	a	t	a
seven	h	i	t	u
hand	r	i	m	a

Hawaiian

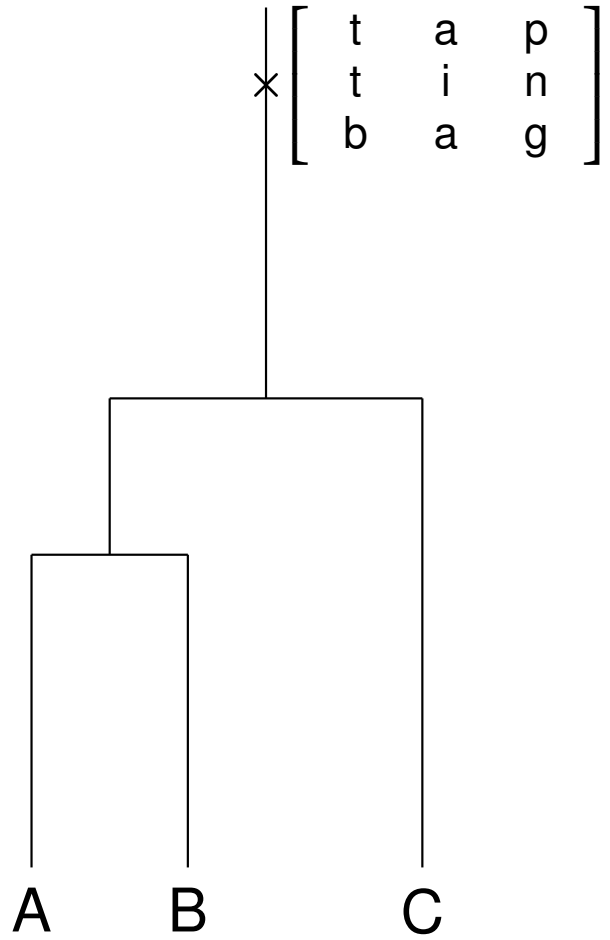
eye	m	a	k	a
seven	h	i	k	u
hand	r	i	m	a

Model: key properties

We need to model the evolution of such a matrix along a tree. We want to allow:

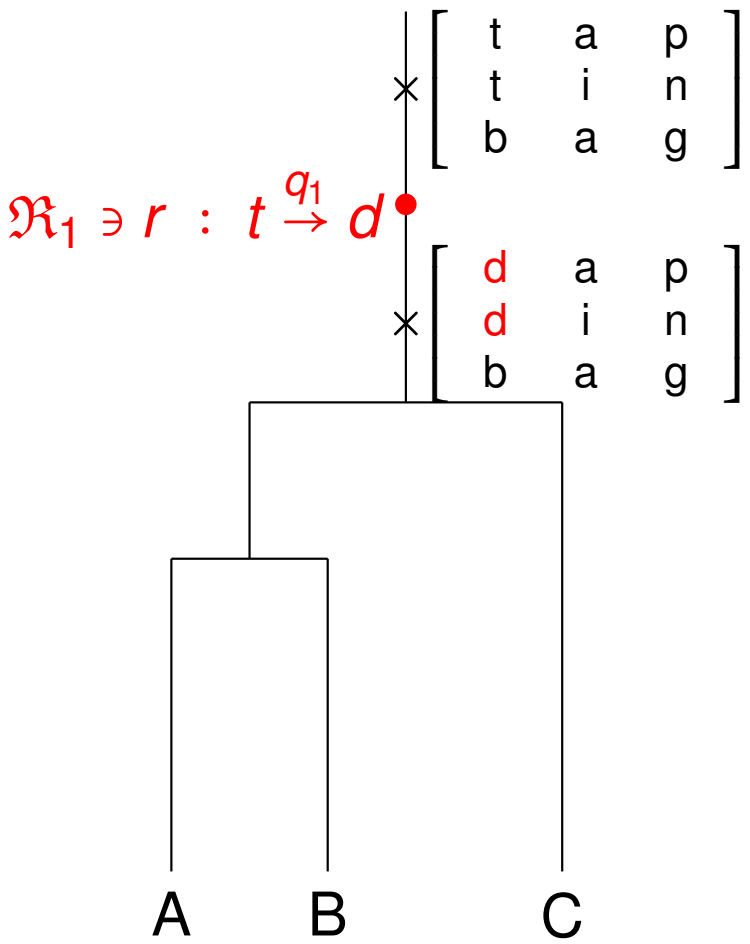
- Apparition of new words: a whole row is modified
- Systematic transformation of a phoneme: within a column, each occurrence of phoneme $/x/$ is changed to phoneme $/y/$ with some probability
- Noise for accidents/unmodelled events

Generative model



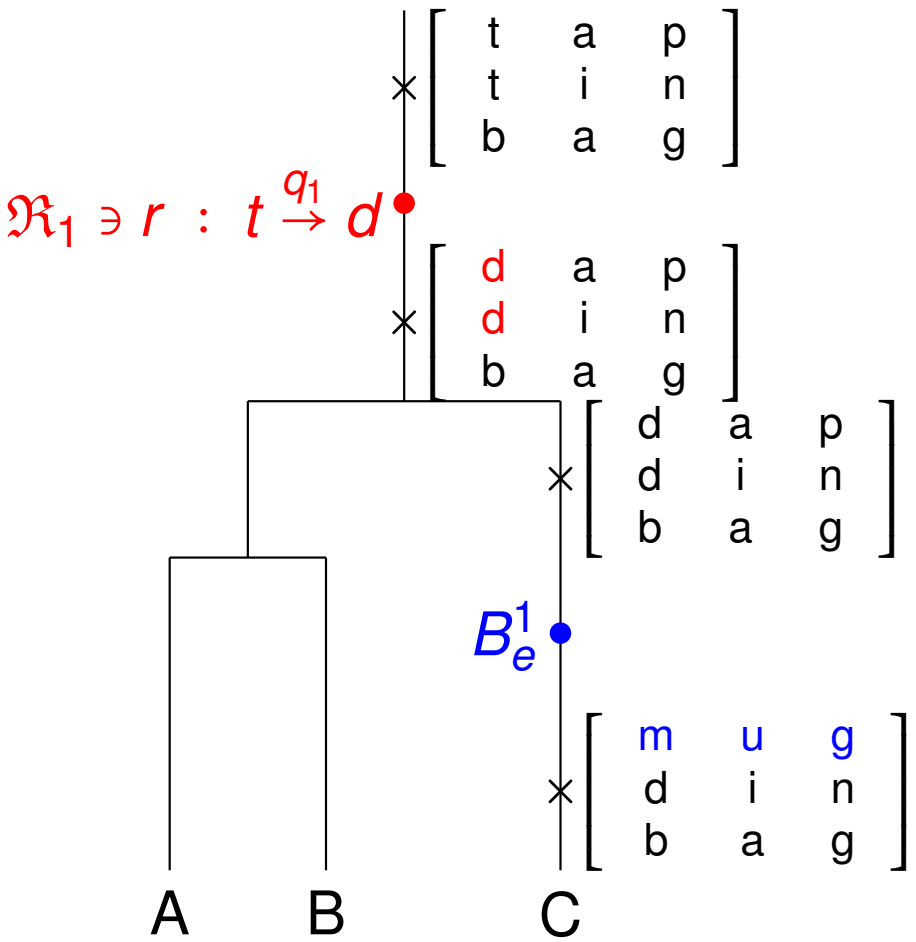
- Binary rooted dated tree, independent process on parallel edges ;
- initial distribution π_0 at the root ;

Generative model



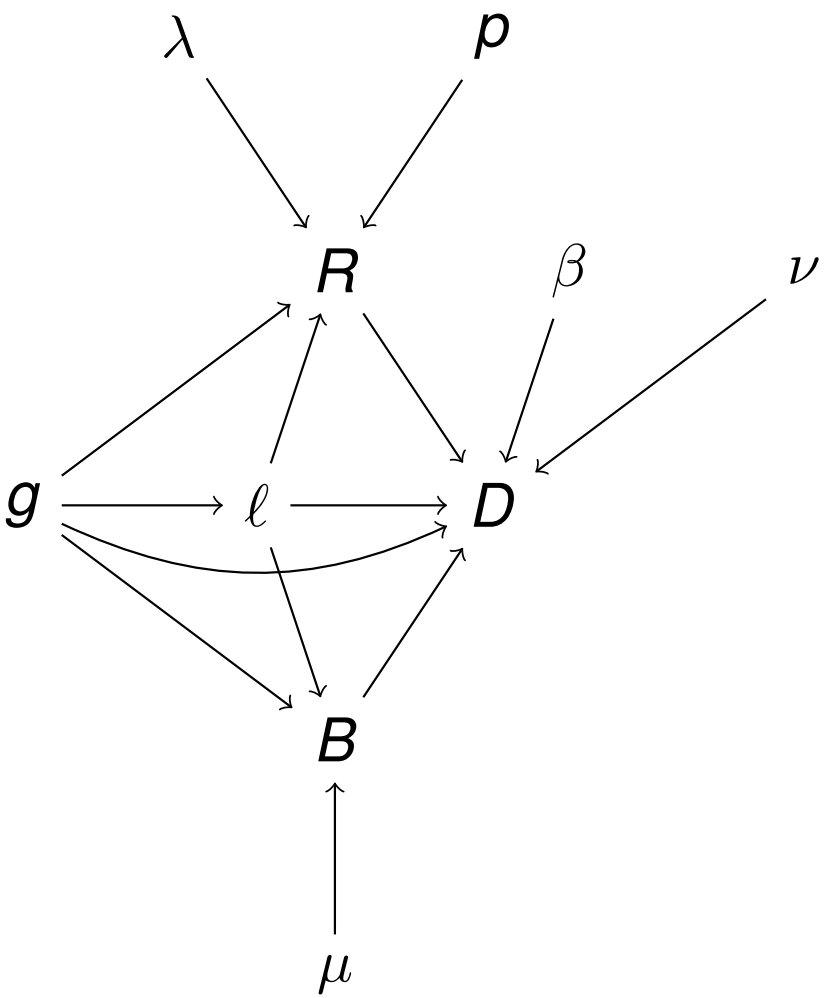
- Binary rooted dated tree, independent process on parallel edges ;
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- random transformation r from a predefined subset $\mathfrak{R}$ of a value of a feature throughout the meanings ;

Generative model



- Binary rooted dated tree, independent process on parallel edges ;
- initial distribution π_0 at the root ;
- random transformation r from a predefined subset $\mathfrak{R}$ of a value of a feature throughout the meanings ;
- random renewal B^m of all the value for a meaning m, also from π_0.

Model dependencies



- Tree g with edge lengths ℓ
- Lexical innovations B occur at rate μ
- Phonological transformations occur at rate λ with probability p, and are applied with probability β
- Noise ν at the end of each edge

Likelihood computation

- The model requires a large number of latent variables (complete lexicon at each internal node + full list of phonological transformations on each edge).
- The likelihood is computable, although expensive. Exploring the posterior is slow.
- Inference procedure: Sequential Monte Carlo, with a long Gibbs sampler at each iteration, and using a tempering scheme on the noise parameter.

Notes

- We are currently able to handle 15-20 leaves on real data.
- The matrix form requires the data to be aligned. This is only true for certain data sets. We cannot handle apparition/disparition of phonemes.
- New misspecifications from this more complete model. In particular, **iconicity**: phonological changes may not be independent of meaning. Tests on synthetic data with iconicity show no bias.

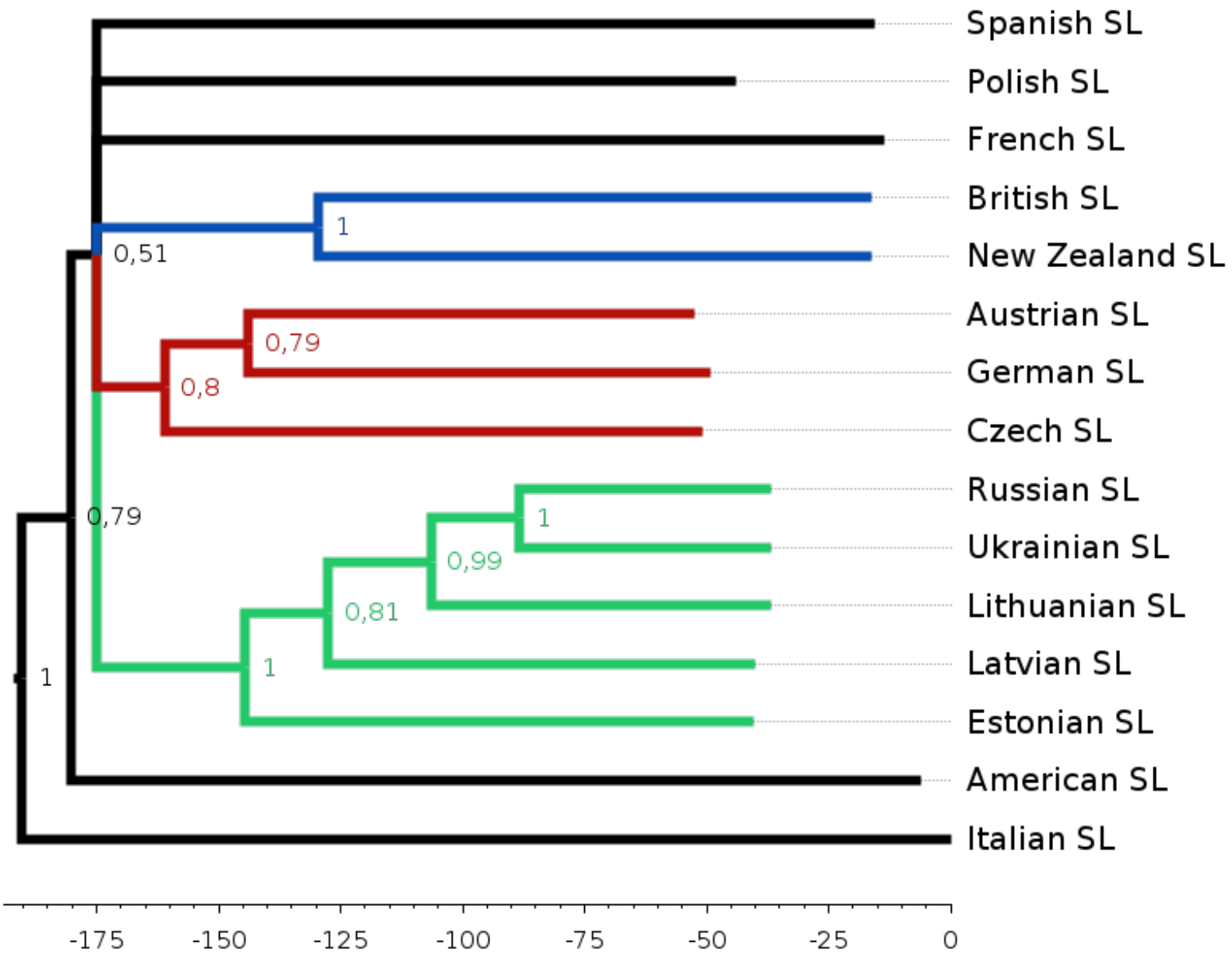
Data from sign languages

Data from 19 sign languages, encoded from video dictionaries.

Italian	Handshape	twohands	handpart
all	5_b	False	back_of_palm
boring	b	False	finger_front
green	5	False	heel_side
Spanish			
all	5	False	ulnar_side
boring	E	False	finger_front
green	V_nb	False	radial_side
Swedish			
all	E	False	ulnar_side
boring	A	True	heel
green	B_nb	False	back_of_palm

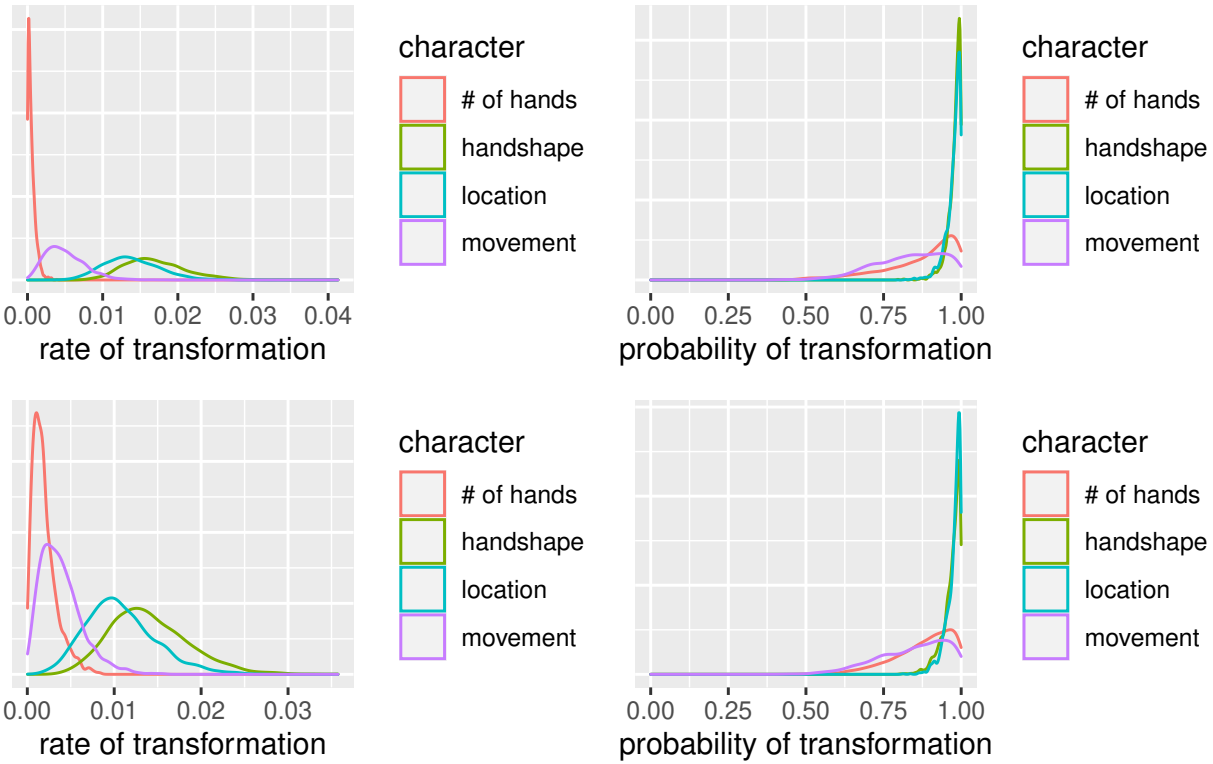
(Modelling) advantage of sign languages: data are aligned; small, known a priori, list of possible phonemes.
We now have a 3D array: languages $\times$ meanings $\times$ characters. Each character belongs to a finite set, mapped to $\{1, 2, \ldots, n_z\}$.

Sign language tree



Abner et al., 2024, Science

Sign language parameters

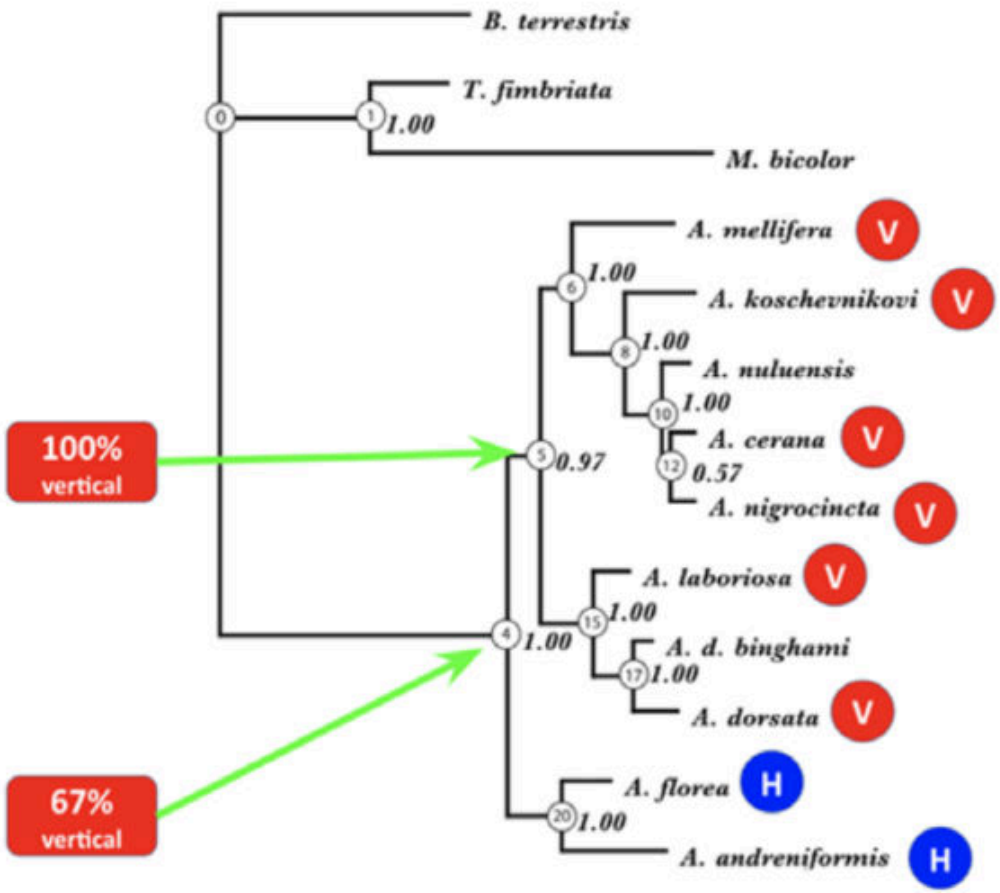


Posterior distributions for the model parameters in for the European SLs dataset [top] and the Asian SLs dataset .[bottom]
Very good model fit: only 15 changes due to noise vs 548 births and 4458 phonological changes. Strong evidence of regular phonological changes.

Outline

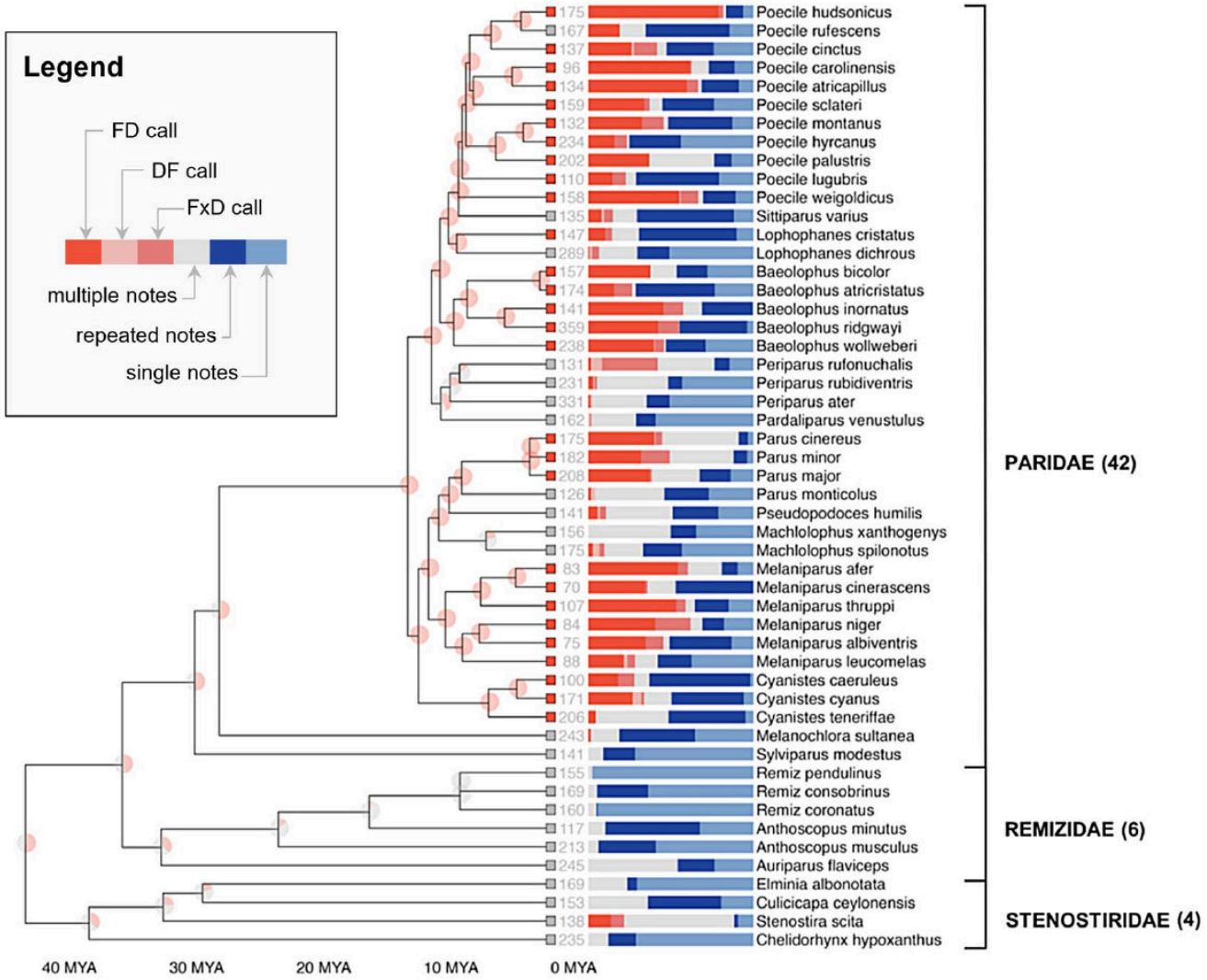
1. Swadesh: Glottochronology
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Bee waggle dances



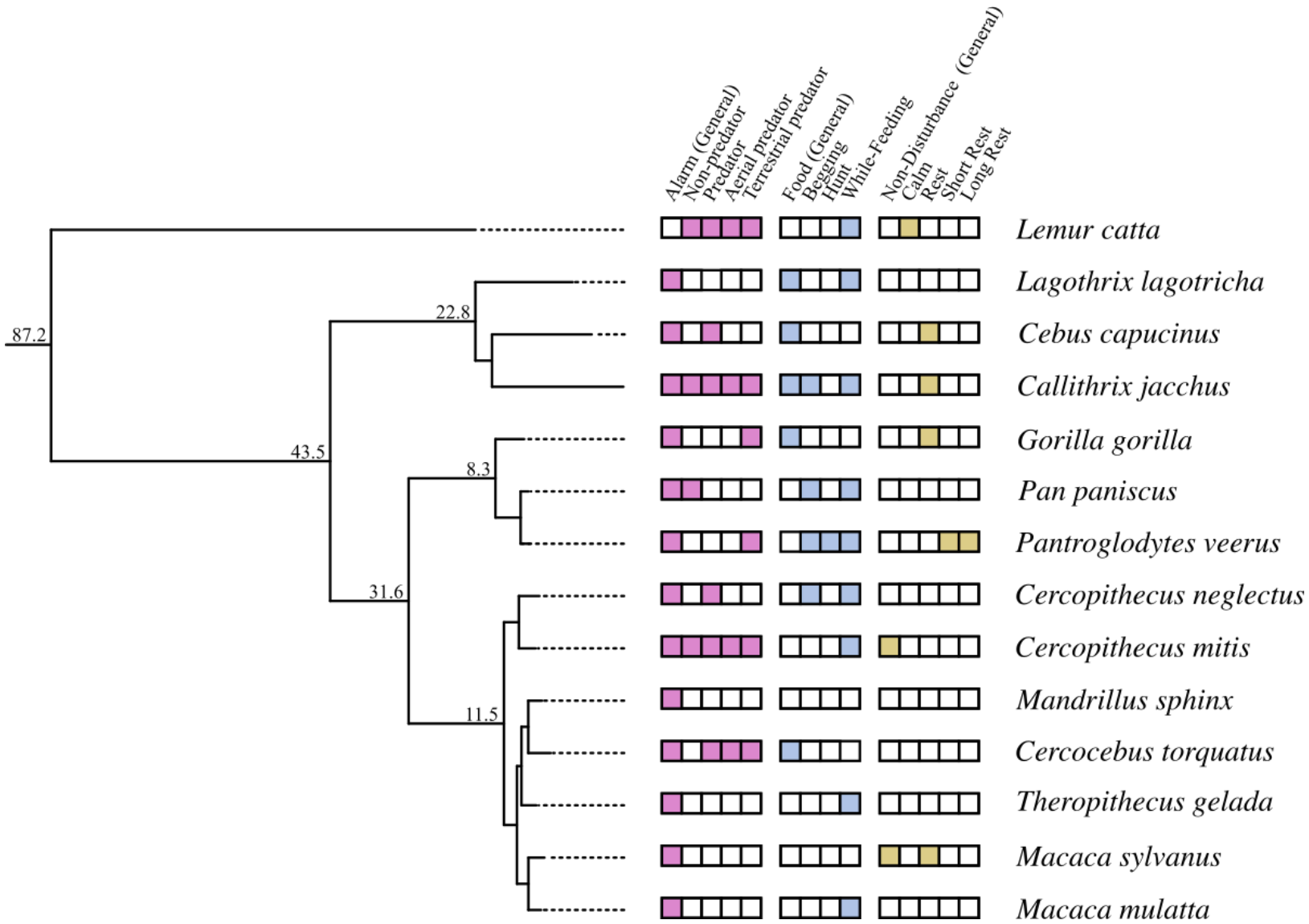
Schlenker et al. 2026, *Ancestral Iconicity*, Biological Reviews

Combination of FD calls in Parids



Salis et al. 2025, *Birds combined calls more than 11 million years ago*, Sc. Rep.

Primate "semantics"



Thouzeau et al. 2026, *Phylogenetics of Primate Call Functions: a methodological proposal and case study*, Proceedings of the COMPAS conference

Questions of interest

- Which (if any) traits are phylogenetically stable?
  - Presence of certain calls?
  - Mappings with contexts of occurrence ("meaning")?
  - Acoustic features?
- Understand the stochastic process underlying changes of these traits
- Reconstruct values for ancestral species

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Overall conclusions

- When done right, statistical methods can provide new insight into linguistic history
- Importance of collaboration in building the model and in checking for mis-specification.
- Bayesian statistics play a big role, for estimating uncertainty, handling complex models and using analyses as building blocks
- Major avenues for future research. Challenges in finding relevant data, building models, and statistical inference:
  - Models for morphosyntactic traits
  - Putting together lexical, phonemic and morphosyntactic traits
  - Semantic change
  - Escape the tree-like model (borrowing, networks, diffusion processes...)
  - Incorporate geography
  - Incorporate evidence from other sources of cultural data

Collaborators



References

- Swadesh, Morris. "Lexico-statistic dating of prehistoric ethnic contacts: with special reference to North American Indians and Eskimos." Proceedings of the American philosophical society (1952): 452-463.
- Gray, Russell D., and Quentin D. Atkinson. "Language-tree divergence times support the Anatolian theory of Indo-European origin." Nature 426.6965 (2003): 435-439. Pagel, Mark, Quentin D. Atkinson, and Andrew Meade. "Frequency of word-use predicts rates of lexical evolution throughout Indo-European history." Nature 449.7163 (2007): 717-720.
- Round, Erich, Rikker Dockum and Robin J. Ryder, "Evolution and Trade-Off Dynamics of Functional Load ", Entropy 24 (2022):507
- Sagart, Laurent, et al. "Dated language phylogenies shed light on the ancestry of Sino-Tibetan." Proceedings of the National Academy of Sciences 116.21 (2019): 10317-10322.
- Clarté, Grégoire, and Robin J. Ryder, "A model of the evolution of discrete matrices for the joint inference of lexical and phonological language histories"

Questions

otázky	kesses
spørgsmåler	cwestiwnau
pytania	preguntes
preguntas	vrae
kláusimai	Fragen
quaestiones	
întrebări	questions
vragen	ερωτήσεις
spurningar	
domande	spørsmåler
questões	frågor
vprašanja	